

2026 DATA ANALYTICS COURSE

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THE BIG DATA ERA

Massive new data streams are opening up a world of opportunities for data scientists!

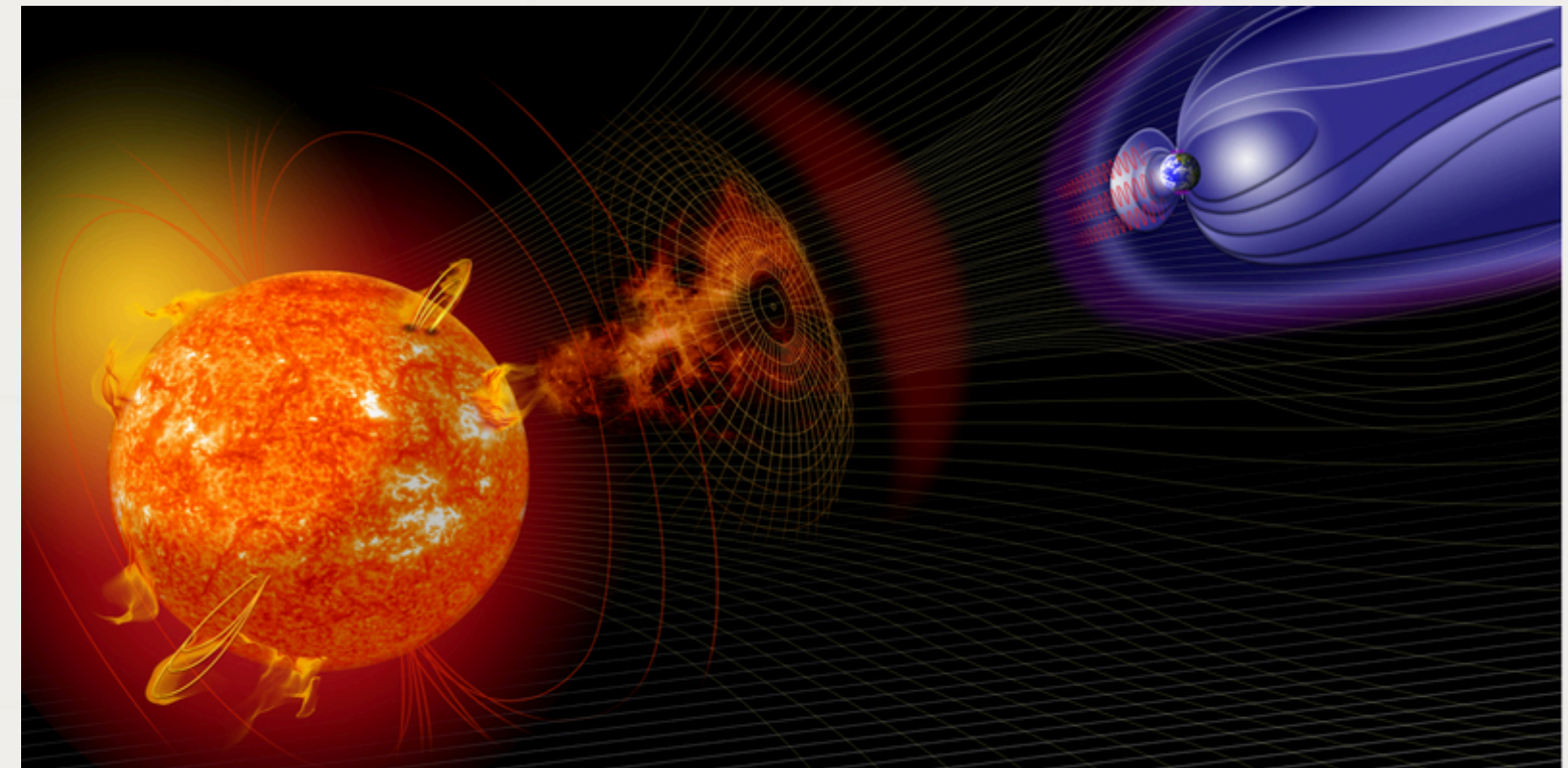
Dramatic increase in the quality and quantity of data:

- massive new surveys: catalogs containing different data,
- high resolution spectrography and imaging across the electromagnetic spectrum,
- massive number of items and/or features,
- space-based telescopes tailored to specific scientific goals, data are not just massive: they are rich, deep, & complex.

THE BIG DATA ERA

Space Weather Effects

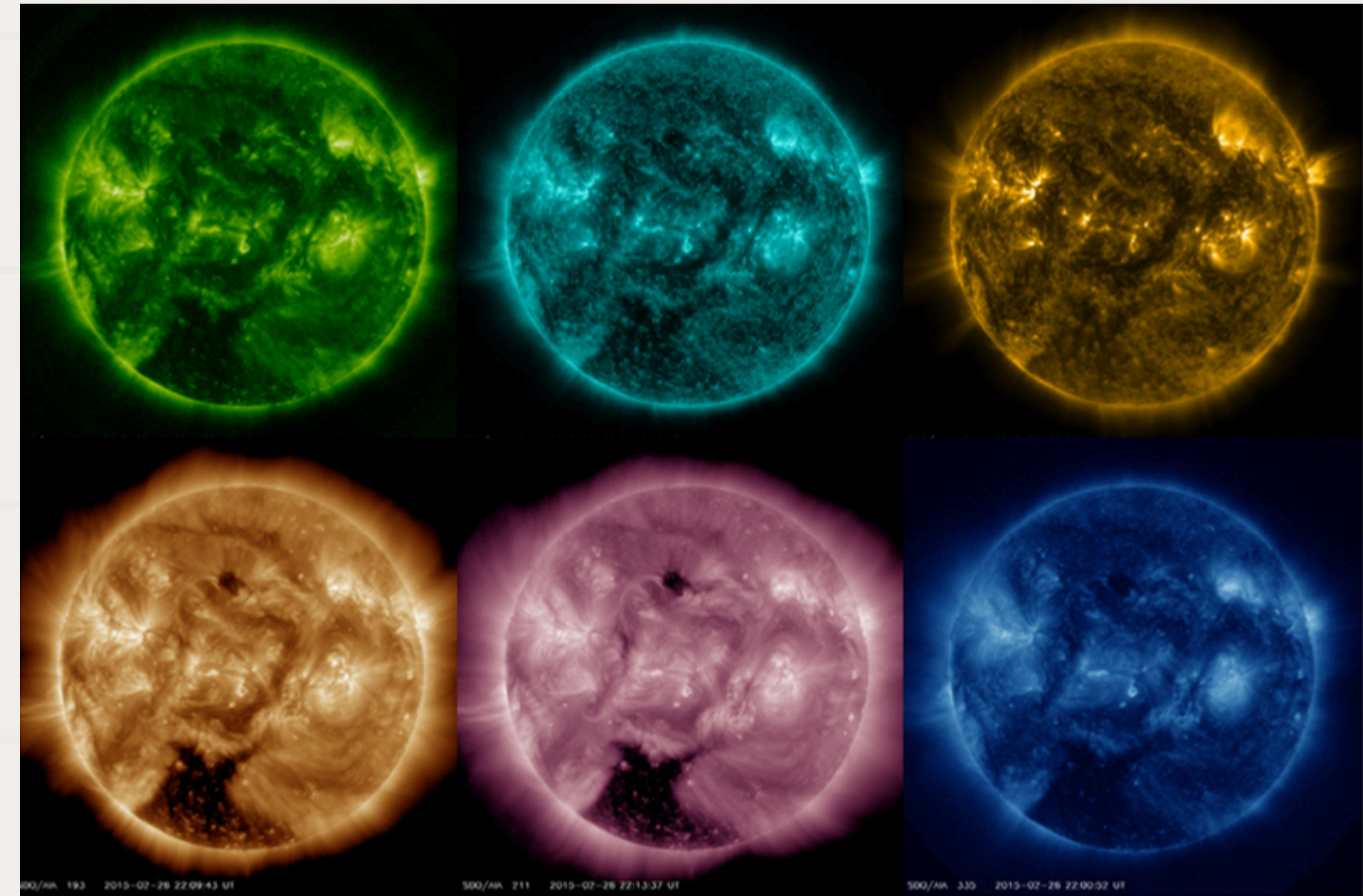
- The highly energetic particles released by solar flares and CMEs can impact the Earth's magnetosphere.
- These impacts cause radio interference and can damage satellites and electric power transmission.
- 1859 geomagnetic storm.



THE BIG DATA ERA

Pixel-by-Pixel Spectra

- The Solar Dynamics Observatory
- NASA satellite launched February 2010
- Massive Data Stream: 1.4 TB/day of compressed data High Spatial and Temporal resolution
- Low Spectral resolution
- White light and magnetigram images



THE BIG DATA REVOLUTION

- Gaia Mission: Cataloging over 1.8 billion stars.
- LSST (Rubin Obs): Will generate ~20 TB of raw data per night starting in 2025.



WHAT IS PROGRAMMING?

A program is a set of instructions written in a specific programming language. These instructions guide a computer to complete tasks step by step. Well-written programs are clear, efficient, and reliable.

PROGRAMMING IS A CRAFT – IT REQUIRES DOING

Computer programming requires practice. If you know a programming language without using it, your knowledge will soon rust away.

We will share with you all the code we use in class. Please use it to clarify your understanding and to practice your skills.

And after the course is over, please continue to study and modify other people's code and then start to write your own. There is no shortcut to writing good programs!





CHALLENGES IN PROGRAMMING

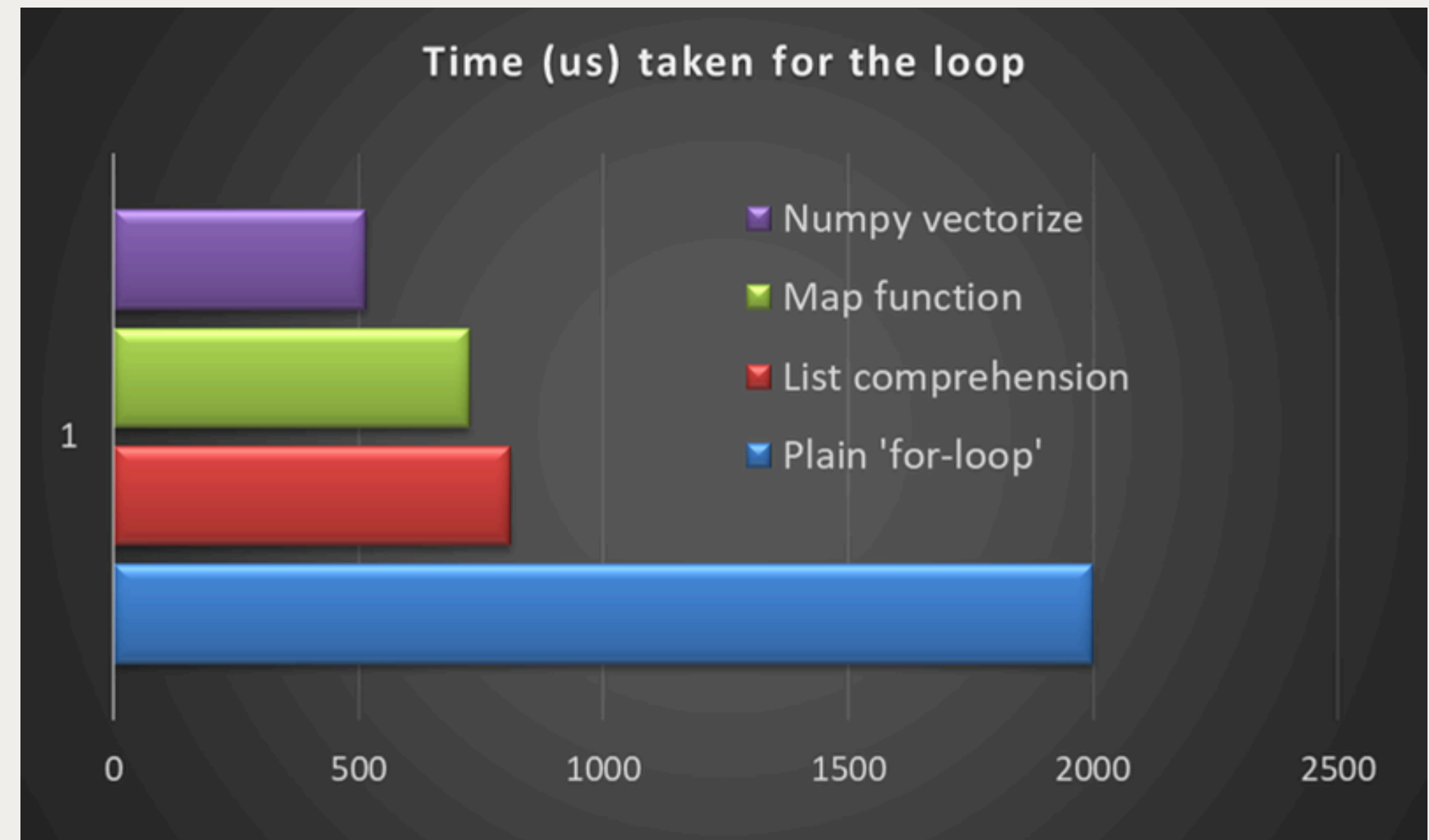
Programming can be challenging due to errors and complex logic. Mistakes are a normal part of learning. Patience and practice help programmers improve over time.



WHY PYTHON



due to its simplicity, extensive scientific libraries, and powerful data analysis capabilities, replacing older proprietary tools like IDL.



INSTALLING PYTHON ON YOUR COMPUTER



Recommended Version

Python 3.10 or 3.11

Download From:

<https://www.python.org/downloads/>



⚠ Important: Do not install Python from the Microsoft Store. The Store version frequently causes issues with:

- venv
- pip
- Jupyter
- "VSCode interpreter detection"

INSTALLING PYTHON ON YOUR COMPUTER



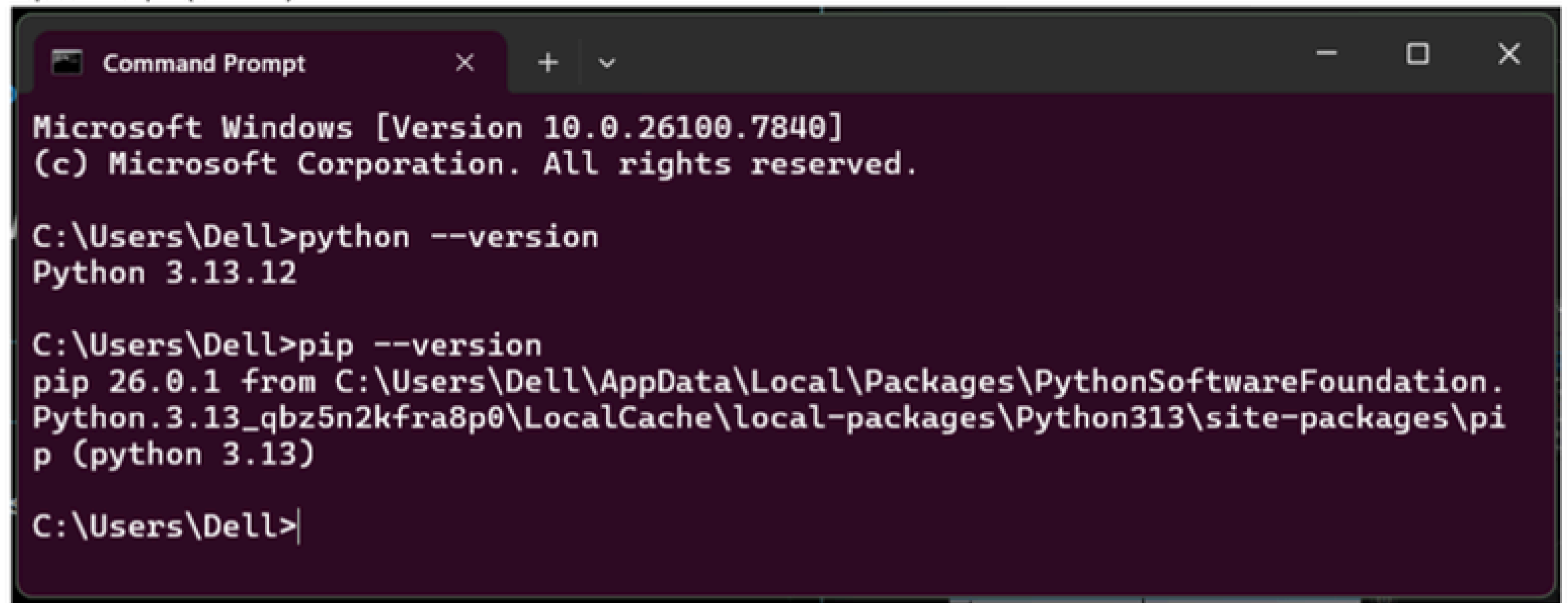
Verify Installation

Open terminal (PowerShell / CMD / Bash):

`python --version`

`pip --version`

Expected output (or similar):



```
Microsoft Windows [Version 10.0.26100.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell>python --version
Python 3.13.12

C:\Users\Dell>pip --version
pip 26.0.1 from C:\Users\Dell\AppData\Local\Packages\PythonSoftwareFoundation.
Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python313\site-packages\pi
p (python 3.13)

C:\Users\Dell>
```


INSTALLING PYTHON ON YOUR COMPUTER

THE ANACONDA DISTRIBUTION

The most recommended method for scientists. It includes almost all necessary packages.

Download installer from anaconda.com



INSTALLING PYTHON ON YOUR COMPUTER

VISUAL STUDIO CODE (VSCODE)



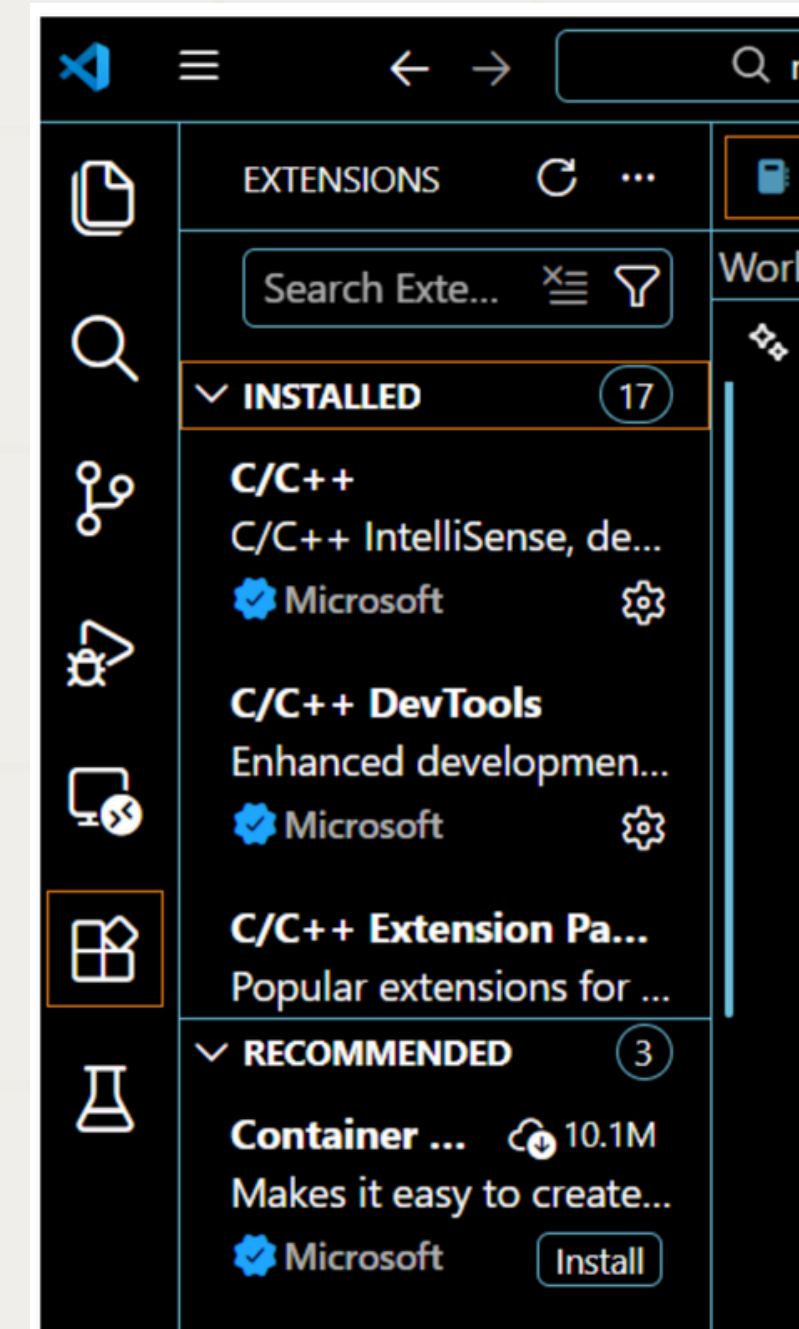
Download from:

<https://code.visualstudio.com/>

Install normally.

Install Required VSCode Extensions

Inside VSCode → Extensions panel.



INSTALLING PYTHON ON YOUR COMPUTER

VISUAL STUDIO CODE (VSCODE)



Required Extensions

1. Python (Microsoft)
2. Jupyter (Microsoft)

Install Required Packages

Upgrade pip first:

```
python -m pip install --upgrade pip
```

Then install core stack:

```
pip install numpy scipy matplotlib pandas scikit-learn jupyter astropy
```



INSTALLING PYTHON ON YOUR COMPUTER

JUPYTER NOTEBOOKS FROM JUPYTERLAB

JupyterLab is an open-source web application that allows you to create and share documents that contain code, plots and narrative text.

The notebooks for each lecture is shared with you.

To install JupyterLab on your computer, please follow the instructions:

https://jupyterlab.readthedocs.io/en/stable/getting_started/installation.htm



COLAB

Installing Python on your computer will give you the maximum and power and flexibility.

However, if you are unable to install it for some reason, you can create online Jupyter notebooks using Google Colab at:

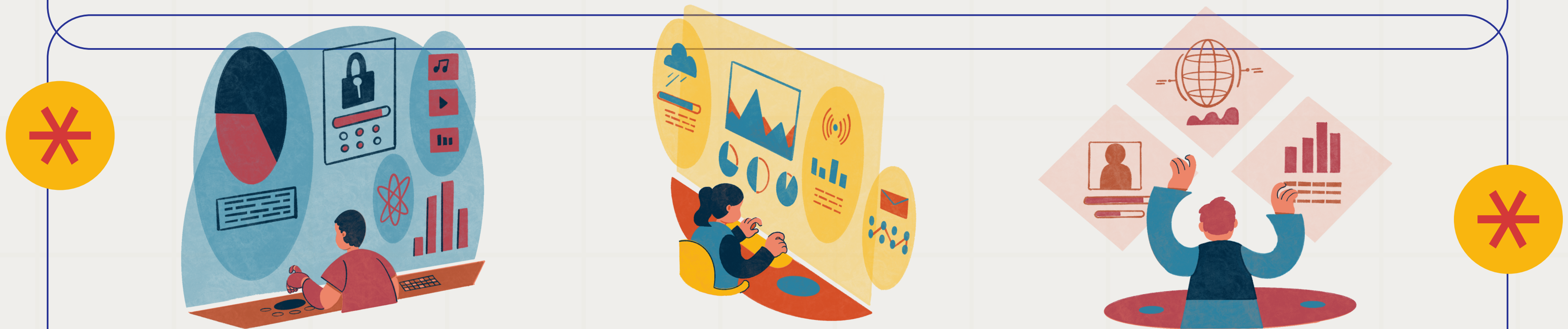
<https://colab.research.google.com/>

You can use Google Drive to save any notebooks you create with Google Collab.

You can even load the notebooks for this course into Google Colab and then modify and run them there.



THANK YOU



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